

Regression and Classification

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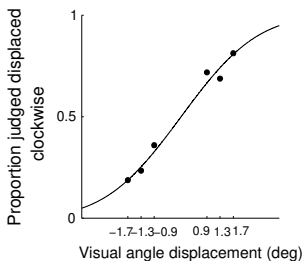
Regression and Classification

Types of variables:

- ▶ quantitative
 - examples: age, height, income
- ▶ qualitative /categorical
 - examples: language spoken, existence of a diagnose

Outcome variables

- ▶ quantitative - regression
- ▶ qualitative - classification
- ▶ But things are not always so clear...



For both regression and classification:

- Explanatory variables can be quantitative or qualitative.
- The focus can be inference or prediction.

Quality of fit - classification

Inference

Error rate is commonly used:

$$\frac{1}{n} \sum_{i=1}^n I(y_i \neq \hat{y}_i)$$

$\hat{y}_i$ is the label predicted by $\hat{f}$

$$I(y_i \neq \hat{y}_i) \begin{cases} 1 & \text{if } y_i \neq \hat{y}_i \\ 0 & \text{if } y_i = \hat{y}_i \end{cases}$$

- ▶ Training error rate should be small.

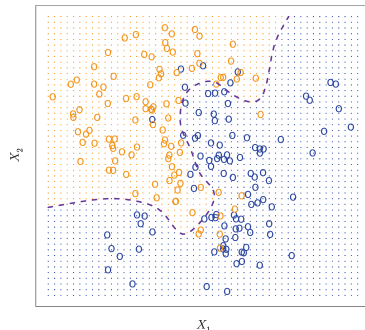


Figure adapted from James, Witten, Hastie, Tibshirani

Prediction:

- ▶ For a new observation x_0 , $\hat{y}_0$ is the correct label.
- ▶ Error rate on test data (test error rate) should be small.
- ▶ $\text{Average}(I(y_i \neq \hat{y}_i))$ should be small.

Bayes classifier

Bayes classifier assigns each observation to the most likely class, i.e. the class j with largest

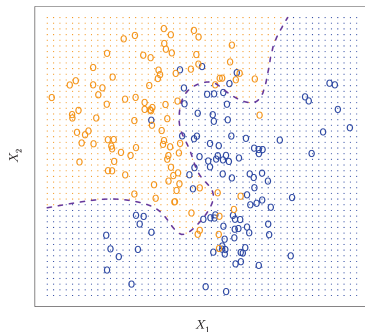
$$\Pr(Y = j|X = x_0)$$

For the example, with 2 classes, for the orange region

$$P(Y = \text{orange}|X) > 0.5$$

simulation: 100 observations in each group

-- Bayes decision boundary



- ▶ The Bayes classifier results in the lowest possible test error rate, the **Bayes error rate**.
- ▶ Analogous to irreducible error in regression.
- The conditional probabilities are unknown for real data!

Classifiers

Discriminative model classifiers: Estimate $\Pr(Y = j|X)$ directly.

- ▶ K-nearest neighbors
- ▶ Logistic regression

Generative model classifiers: Estimate $\Pr(Y)$ and $\Pr(X|Y)$ to calculate $\Pr(Y = j|X)$ using Bayes theorem.

- ▶ Linear discriminant analysis
- ▶ Quadratic discriminant analysis

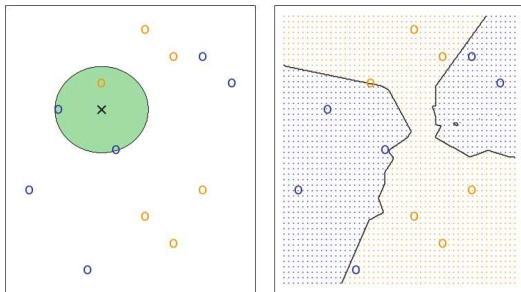
Discriminant classifiers: Find a discriminant function that directly maps an input x to a category C_k .

- ▶ Support vector machines

K-nearest neighbors classifier

- ▶ For a given K and a test observation x_0 , the K points in the training set closest to x_0 are selected ($\mathcal{N}_0$) and the conditional probabilities are estimated as:

$$\Pr(Y = j|X = x_0) = \frac{1}{k} \sum_{i \in \mathcal{N}_0} I(y_i = j)$$

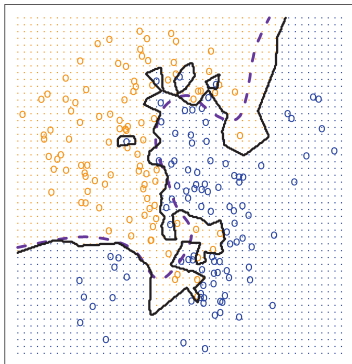


- ▶ Bayes rule is applied to classify x_0 to the class with largest probability.

Overfitting and bias-variance trade off

Overfitting and bias-variance trade-off also apply to classification problems.

$K = 1$



$K = 100$

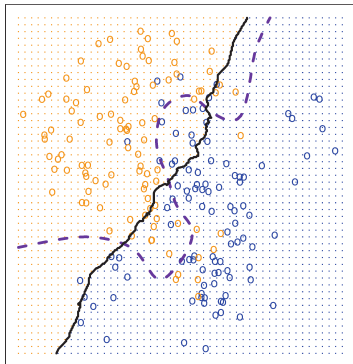
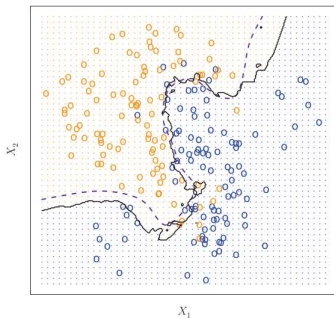


Figure adapted from James, Witten, Hastie, Tibshirani

Overfitting and bias-variance trade off

$K = 10$



-- Bayes error rate

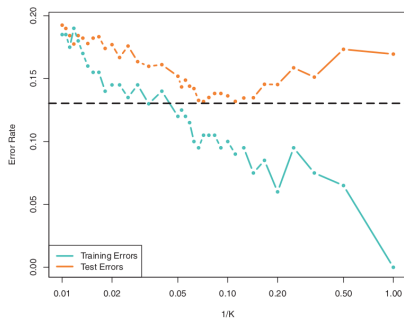
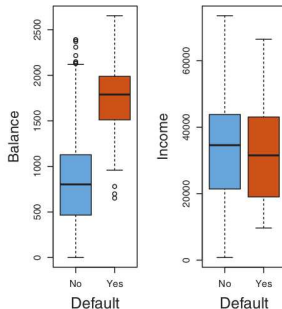
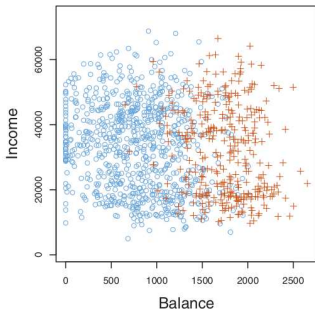


Figure adapted from James, Witten, Hastie, Tibshirani

Regression for a categorical output variable



$$Y = \begin{cases} 0 & \text{if No} \\ 1 & \text{if Yes} \end{cases}$$

$$y \sim \text{Bernoulli}(p)$$

$$\text{Model: } \Pr(Y = 1 | X = x_0) = p(X)$$

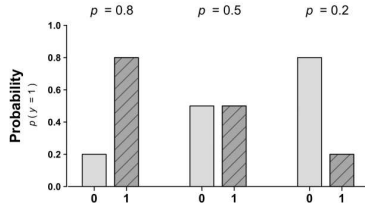


Figure from Winter

Logistic regression model

The linear model $p(X) = \beta_0 + \beta_1 X$ would result in probabilities smaller than 0 and larger than 1!

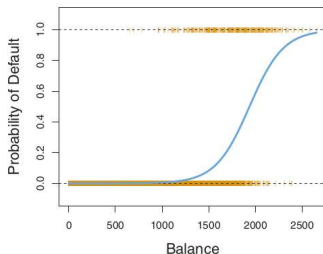
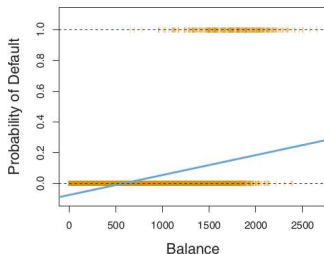


Figure adapted from James, Witten, Hastie, Tibshirani

The log of the odds
 $p(X)/(1 - p(X))$ is modeled
linearly:

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X$$

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

Odds

Odds can take values between 0 and ∞ .

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X$$

$$\frac{p(X)}{1-p(X)} = e^{\beta_0 + \beta_1 X}$$

Example:

- ▶ $p(X) = 0.2$ on average 2 out of 10 people default, means that the odds for defaulting are $\frac{0.2}{1-0.2} = 1/4$
- ▶ $p(X) = 0.9$ on average nine out of 10 people default, means that the odds for defaulting are $\frac{0.9}{1-0.9} = 9$

Logistic regression

- ▶ Logistic regression can be used for **inference** about the importance of different variables to the outcome or for **prediction**.
- ▶ Logistic regression can be used with several explanatory variable (Multiple Logistic Regression).

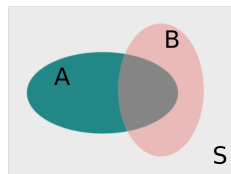
$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

- ▶ Multinomial logistic regression or Softmax regression (more than 2 categories).

Conditional probabilities and Bayes theorem

Generative model classifiers use Bayes theorem to estimate $\Pr(Y = j|X)$.

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$



$$\Pr(B|A) = \frac{\Pr(A|B) \cdot \Pr(B)}{\Pr(A)}$$

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)} \quad (1)$$

$$\Pr(B|A) = \frac{\Pr(A \cap B)}{\Pr(A)} \quad (2)$$

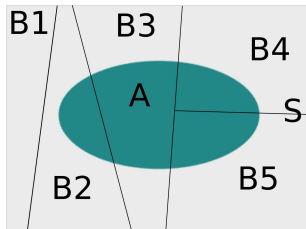
One rewrites (1) to get $\Pr(A \cap B) = \Pr(A|B) \Pr(B)$.

Replacing this in (2) one gets Bayes theorem.

Bayes Theorem

Using the Law of Total Probability:

$$\Pr(B_i|A) = \frac{\Pr(A|B_i) \cdot \Pr(B_i)}{\sum_{j=1}^k \Pr(A|B_j) \cdot \Pr(B_j)}$$



Estimating $p_k(x)$ (posterior) using Bayes theorem:

$$\Pr(Y = K|X = x) = p_k(X) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$$

- ▶ π_k estimation is simple: fraction of the training observations that belong to the k th class
- ▶ $f_k(x)$ estimation assuming normality

Linear Discriminant Analysis

For the case with one explanatory variable X , assuming:

- ▶ observations within each class come from a normal distribution with class specific mean μ_k ,
- ▶ all classes have a common variance σ^2 .
- ▶ Assign X to the class with largest $\delta_k(x)$

$$\delta_k(x) = x \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k) \quad \hat{\pi}_k = n_k/n$$

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i \quad \hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$$

- Assuming class specific variances σ_k^2 - Quadratic Discriminant Analysis.

Linear Discriminant Analysis - $p = 1, K = 2$

Example with $p = 1, K = 2$ and $\pi_1 = \pi_2$:

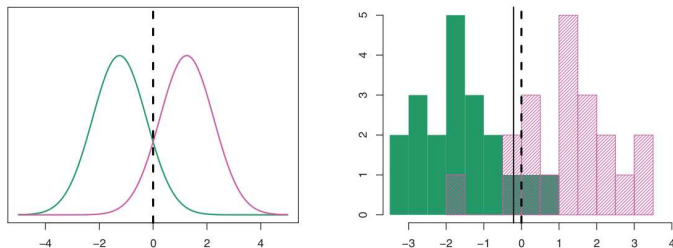
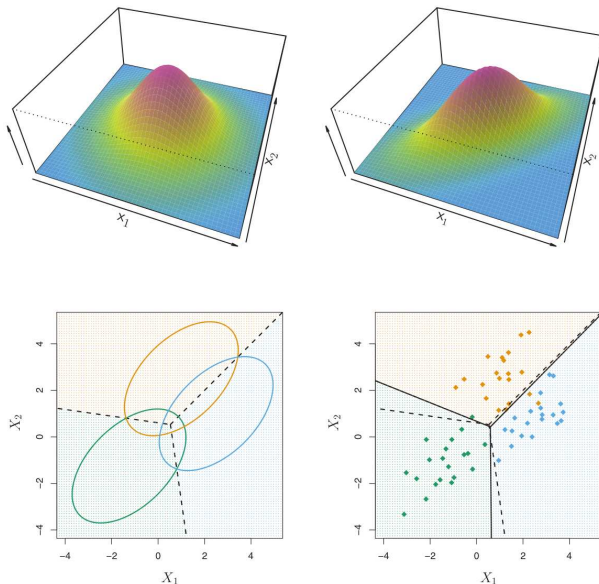


Figure adapted from James, Witten, Hastie, Tibshirani

- ▶ 20 observations from each class
- ▶ solid line: LDA decision boundary estimated from training data
- ▶ dashed line: Bayes decision boundary $(\mu_1 + \mu_2)/2$

Linear Discriminant Analysis - $p > 1$, $K > 2$

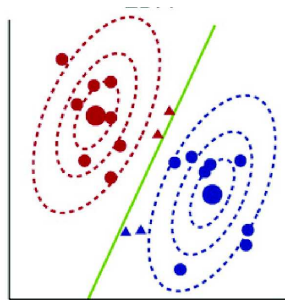
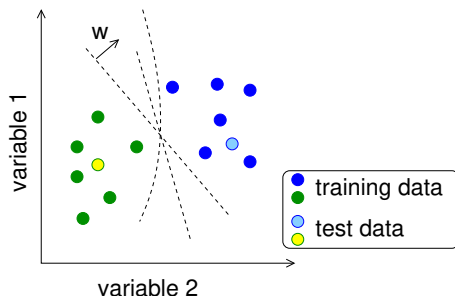


Figures from James, Witten, Hastie, Tibshirani

Geometric interpretation

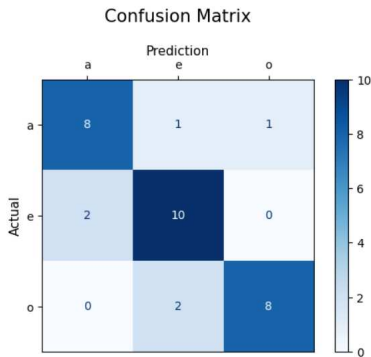
Linear classification:

- ▶ an hyperplane is defined by a weight vector
- ▶ the weight vector indicates the importance of each variable



Confusion matrix

It is interesting to determine what type of errors are made:

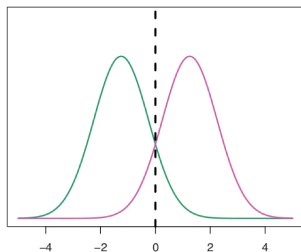


	Predicted a	Predicted e	Predicted o
a	8	1	1
e	2	10	0
o	0	2	8

- For a binary classification with one small class, a null classifier will have low error rate!

Types of error

	$Y = 0$	$Y = 1$
predict $Y = 0$	True Negatives Specificity	False negatives
predict $Y = 1$	False Positives	True Positives Sensitivity



For 2 classes:

$$\Pr(Y = j|X = x_0) > 0.5$$

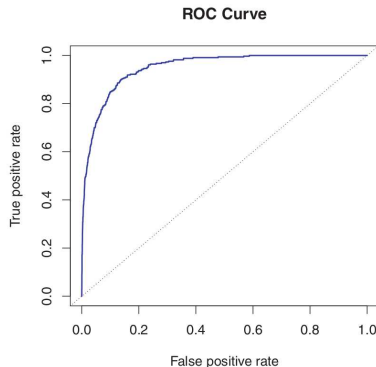
ROC curve

$$\Pr(\text{default=yes} | X = x) > 0.5$$

		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9,644	252	9,896
	Yes	23	81	104
Total		9,667	333	10,000

$$\Pr(\text{default=yes} | X = x) > 0.2$$

		<i>True default status</i>		
		No	Yes	Total
<i>Predicted default status</i>	No	9,432	138	9,570
	Yes	235	195	430
Total		9,667	333	10,000



- ▶ The ROC curve summarizes performance of a classifier over all thresholds.
- ▶ Better classifiers gave larger the area under the ROC curve.
- ▶ The ROC curve is useful to compare classifiers.